

Utkarsh Solanki

Cybersecurity Student | Web Application Security & Penetration Testing

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SUMMARY

Final-year BCA student focused on cybersecurity, with hands-on experience in web application security and penetration testing through self-directed labs. Completed 85+ TryHackMe rooms and 34 PortSwigger Web Security Academy labs, and built Python-based security tooling — an HTTP proxy, a directory brute-forcer, and a penetration-testing toolkit — to understand vulnerabilities at the protocol level. Seeking a cybersecurity internship in web application security / penetration testing.

EDUCATION

Bachelor of Computer Applications (BCA) — Galgotias University

Final-Year Student

TECHNICAL SKILLS

- **Security:** Web Application Security, Penetration Testing, OWASP Top 10, SQL Injection (UNION-based, blind, error-based), Cross-Site Scripting (XSS), Access Control Vulnerabilities, SSRF, Path Traversal
- **Tools:** Burp Suite, Nmap, Metasploit Framework, Hydra, Netcat, Git/GitHub
- **Networking:** TCP/IP, HTTP, DNS, Sockets, Subnetting
- **Programming:** Python (security tooling, sockets, threading, automation), Bash
- **Platforms:** Linux (Kali Linux), Metasploitable 2, DVWA, WebGoat, TryHackMe, PortSwigger Web Security Academy

PROJECTS — PYTHON SECURITY TOOLING

HTTP Proxy & Parser — `http-proxy-lab`

Python (Standard Library)

- Built a forward HTTP proxy and request/response parser from raw TCP sockets (no frameworks) to study HTTP internals end-to-end.
- Implemented Content-Length-driven body reads and origin-form path rewriting; documented the proxy's own attack surface (open-proxy/SSRF exposure, unbounded-buffer DoS).

pentools — Python Security Toolkit

Python, requests, pypdf

- Built five standalone CLI security tools: a TCP port scanner, hash identifier/cracker, password hasher (MD5–SHA512), PDF password auditor, and blind SQL injection extractor.
- The blind SQLi tool automates a PortSwigger lab exploit (Oracle conditional-error oracle), extracting data character-by-character via HTTP response codes.

dir-brute — Directory Brute-Forcer & Crawler

Python, threading, argparse

- Built a multithreaded content-discovery tool that brute-forces hidden directories from a wordlist, filters by HTTP status code, and supports recursive re-scanning.
- Added URL normalization and a crawl mode; validated end-to-end in a documented lab writeup against a local test server.

HANDS-ON CYBERSECURITY EXPERIENCE

PortSwigger Web Security Academy — 34 Labs Solved

- Solved 34 labs across five vulnerability classes: Access Control (13/13), SQL Injection (15/18 — UNION, blind, and error-based, across MySQL, Oracle & MSSQL), Path Traversal, XSS, and SSRF.
- Documented each lab with recon, exploitation steps, payloads, root cause, and mitigation.

Metasploitable 2 / DVWA / WebGoat — 10 Documented Labs

- Exploited 4 CVEs on Metasploitable 2 with the Metasploit Framework (vsFTPD backdoor, PHP CGI argument injection, Samba usermap_script, UnrealIRCd backdoor), plus SSH and Telnet credential brute-forcing.
- Performed a Hydra-driven brute-force and a blacklist-bypass reflected-XSS attack on DVWA, and an SSRF parameter-tampering attack on WebGoat via Burp Suite.

TryHackMe — 85+ Rooms, SEC0 Certified

- Completed the Pre-Security path (SEC0 Professional Certification) and Cyber Security 101 (14/14 modules); currently working through the Jr Penetration Tester path.
- Authored structured writeups (commands, concepts, tools) for 85+ rooms across networking, Linux, cryptography, exploitation, and web security.